Examiner only

5. Springs are used in a wide variety of applications, ranging from those used in the suspension systems of heavy vehicles to those used in retracting ball-point pens.

The table below shows the properties of a number of springs that are available from a particular supplier.

Spring	Coil diameter (mm)	Wire diameter (mm)	Maximum load (N)	Unstretched length (mm)	Spring constant (N/mm)	Maximum length (mm)
A	16.00	0.165	1.252	6.350	0.17	13.716
B	16.00	0.165	1.031	6.350	0.14	13.716
C	16.00	0.165		6.350	0.12	17.050
D	16.00	0.165	1.257	9.524	0.09	23.495
E	16.00	0.230	1.768	7.950	0.24	14.016
F	16.00	0.230	1.524	7.950	0.12	20.650

(a) (i) Write down the spring that can withstand a maximum load of at least 1.2 N and has a maximum extension close to 14 mm. [1]

Spring = ...

(ii) Express the spring constant for **spring A** in standard units of N/m. [1]

Spring constant, k = ... N/m

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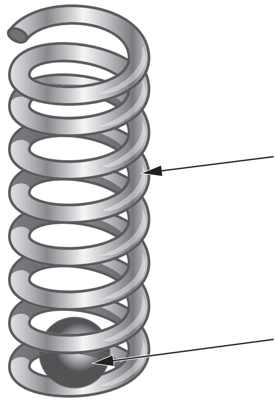
(b) A group of students want to use a spring to project a small ball vertically upwards. They choose to use **spring C**. The bottom of their spring would hold the ball within it and the top of the spring would be fixed in position.
The bottom of the spring would be pulled down and released to project the ball vertically upwards.



(i) Use equations from page 2 to calculate the energy stored in the spring when it is stretched to its maximum length. [3]

Stored energy = ... J

(ii) Use an equation from page 2 to calculate the maximum theoretical height that the ball would reach given that its mass is 15 grams. [3]
(g = 10 N/kg)

Height = ... m

THIS QUESTION CONTINUES ON PAGE 16

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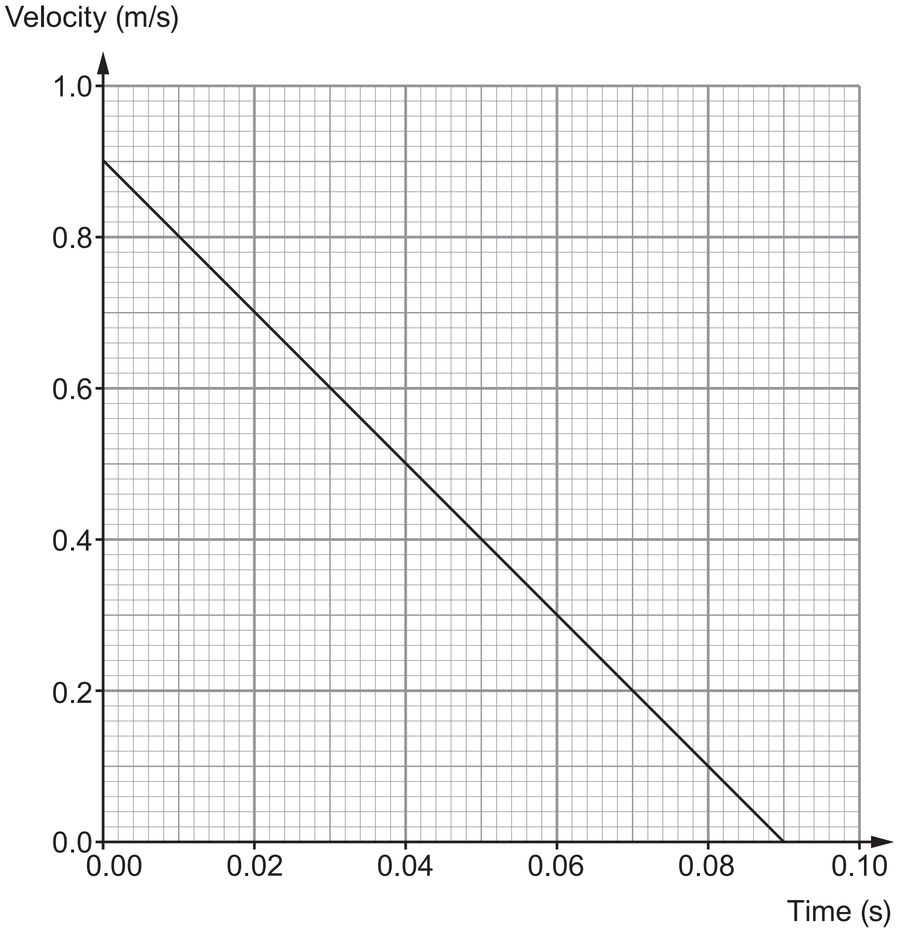
(iii) A data logger is then used to monitor the motion of the ball as it rises from the spring. A graph is automatically produced showing how the velocity of the ball changes with time. The graph is shown below.



I. One student states that the height reached by the ball is different from that which you have calculated in *(b)*(ii). Show that this statement is true. [2]
Space for calculation.

....

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II. Explain why the height reached by the ball is different from the theoretical height. [2]

....

....

....

III. Suggest **one** change that could be made that would enable a ball to reach a greater height in this experiment. [1]

....

....

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END OF PAPER